
Hot Peppers

There is a wide variety of hot peppers, and their capsaicin content (the chemical compound that causes the burning sensation) varies from one cultivar to another. The choice of cultivars therefore depends on our market. The selection is made based on which types of hot peppers are likely to interest our customers. In terms of production management, the pepper is very similar to the field bell pepper.

Family

- Solanaceae

Cultivars

- Baron (Poblano)
- Espelette
- Jedi (Jalapeno)
- Red Rocket (Cayenne)
- Habanero
- Mellow Star (Shishito)
- Caperrino (Cherry)

Crop Planning

We plan only one hot pepper planting per season in our calendar. As this vegetable is very sensitive to cold, we try to plant it when temperatures are higher and prefer to grow it under shelter (caterpillar tunnel, unheated greenhouse).

Once transplanted, it takes 6 to 12 weeks, depending on the variety, before the first fruit is harvested. The number of days to maturity can also be taken into account when selecting varieties. For example, once transplanted, the shishito pepper, which is harvested green, requires about 45 days before harvesting

begins, while the habanero pepper, whose fruit is harvested colored, requires 80 days.

Starting Seeds

- Start seedlings 7-8 weeks before transplanting date. Sow in 128-cell seed trays filled with a low fertility potting mix. 1 seed per cell.
- Place the trays on a heat mat and maintain the temperature at 80°F (27°C) until germination.
- After germination, keep the nursery room temperature around 75°F (24°C) during the day and 65°F (18°C) at night. The soil must be slightly humid at all times.
- 3 weeks after sowing, repot in 4 in (10 cm) pots using a fertile potting mix.
- Plants are ready to be transplanted when they reach a height of 6-8 in (15-20 cm) and unopened flower buds are present.
- 1 week before transplanting, harden off the plants by exposing them to outdoor temperatures and cutting back on water.

Intensive Spacing

2 rows per bed, with an in-row spacing of 12 in (30 cm).

Fertilization

- Hot peppers are a very demanding vegetable in terms of fertilization and remain in the field for more than 100 days. Provide full nutrient requirements when preparing the bed.
- Apply plenty of compost. A layer of about 2 in (5 cm) should be spread evenly over the bed. Add a nitrogen-rich amendment such as a mixture

of pelleted chicken manure (5-3-2) and alfalfa meal (2-0-2). Provide an application of potassium sulfate (0-0-50) and organic Sul-Po-Mag (0-0-22).

Irrigation

Irrigation System

- Drip tapes (4 lines per bed): under landscape fabric.

Irrigation Management

- Root depth: medium (24-40 in [60-100 cm]).
- General information: ensure uniform and constant watering from flowering stage to fruiting. Peppers like the soil to dry out slightly between watering.
- In hot weather, an imbalance in watering can cause blossom end rot, a sign of calcium deficiency. Frequent and thorough watering helps prevent this problem.

Implantation

Hot peppers like the soil to be at least 60°F (16°C), and ideally 68°F (20°C). If necessary, install clear plastic over the beds 2 weeks before the transplant date to warm the soil. A tunnel also helps to warm the soil more quickly.

Equipment

- Broadfork
- Fertilization and amendments
- 30 in (75 cm) bed prep rake
- Power harrow
- Irrigation equipment (drip irrigation)

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- Landscape fabric and anchoring pins

Itinerary

- Before transplanting, make sure the soil is warm enough.
- To make transplanting easier, we prepare each bed just before planting (since the soil is very loose after being harrowed).
- Remove the tarp and quickly rake the bed if there is a lot of large debris.
- Prepare each bed with the broadfork.
- Apply fertilizer and amendments evenly, then power harrow to a depth of 2 in (5 cm).
- Install 4 drip lines per bed (2 per row) and start the system. Repair any leaks if necessary.
- Lay down the landscape fabric and secure it with anchoring pins at every second step (5 ft or 1.5 m). It should be very tight. Make sure the drip lines underneath are close to the transplant holes.
- Bring the well irrigated pots near your bed. Unlike many other crops, it is recommended that you take your time to transplant this one properly. Dig a hole with a trowel. Place the plant at the bottom of the hole and fill the spaces on the sides with excess soil. Make sure that the top of the root ball is level with the soil, then press firmly.
- Irrigate the soil thoroughly once, then keep the soil moist for a week after planting.

Weeding and Maintenance

- Weed by hand, 2 or 3 times during the season, inside the landscape fabric holes.

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- Using a sledgehammer (or a stake-driver), drive the 4 ft (1.2 m) steel rods (rebar) about 12 in (30 cm) into the ground. Place one for every 5 plants. Once the plants have reached about 16 in (40 cm) in height, install the first line of bale rope by passing it from one side of each stake to the other. Depending on the cultivar, repeat this procedure once more as soon as the plants grow an additional 8 in (20 cm).

Plant Protection

- Young pepper plants can be attacked by aphids. To solve this problem, remove seriously affected leaves and introduce beneficial insects (ladybugs or *Aphidius colemani* parasitic wasps). If an infestation is present, treat the plants with a biopesticide approved for this purpose. Wait 24 to 48 hours before introducing insects.
- If plants bloom, but do not fruit, this is usually a sign of tarnished plant bug (at least in more northern areas), which is very fond of the sap of eggplant and pepper flowers. In the case of a heavy infestation, it may be necessary to treat with a biopesticide approved for this purpose.
- Blossom end rot is a physiological disorder generally related to calcium deficiency, which in turn is related to irrigation and humidity. To control blossom end rot, water thoroughly during the fruit formation period, and especially during the fruit ripening period, and provide adequate ventilation. Aim for up to 1 in (2.5 cm) of water per day (including rain). Also, in hot weather or when blossom end rot is present, spraying with calcium chloride may help slow the problem.

Harvests

- Harvest the peppers twice a week. The harvest period is usually 6 to 10 weeks depending on the variety grown.
- Pick the peppers using your hands (remember to wear gloves) and tree planting bags. Harvest the plant from the bottom to the top.

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- It is very important to never mix cultivars when harvesting. Use different harvest crates for each cultivar.
 - For some varieties, the fruit is harvested when it is still green (poblano, shishito, jalapeno), but for the other varieties, we want the fruit to be ripe and to have reached its full color.
 - Only peppers free of blemishes are harvested. The remaining peppers must still be picked, but they are left in the aisle, except for peppers with blossom end rot, which can be left on the plants. This practice reduces the risk of rotting other fruits.

Conditioning

- The peppers are not washed, but should be wiped clean if they are dirty.

Storage

- Peppers can be kept in a cool room at 46-54°F (8-12°C) and should not be kept longer than 4 days before being sold.